

Kowshik S

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EDUCATION

College of Engineering, Guindy

Chennai, Tamil Nadu

Bachelor of Engineering in Computer Science and Engineering – CGPA: 7.73/10

2023 – 2027 (Expected)

- **Relevant Coursework:** Data Structures, Design and Analysis of Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Full Stack Technologies, Machine Learning

Vidyaa Vikas Matric Higher Secondary School

Coimbatore, Tamil Nadu

Class XII, Matriculation Board – 97.5% (585/600)

2022 – 2023

- School First across all Vidyaa Vikas institutions in Tamil Nadu

EXPERIENCE

Department of Computer Science & Engineering, Anna University, CEG

Chennai, Tamil Nadu

Web Development Intern – JWT-Based Login System for Student-Faculty Portal

May 2026 – Jun 2026

- Engineered a role-based authentication system (separate student/faculty login flows) using JWT and bcrypt, with full-stack features built on the MERN stack.
- Implemented device fingerprinting to validate that each API request originates from the same device/browser as the original login, preventing a copied or stolen token from being reused elsewhere.

PROJECTS

URL Shortener

Jul 2026

- Built a full-stack URL shortener with JWT-based stateless authentication and bcrypt password hashing, enforcing per-user ownership so users can only manage links they created.
- Designed a Redis cache-aside layer in front of PostgreSQL for link redirects and per-user/IP rate limiting, and logged click analytics (count, referrer, user-agent, timestamp) asynchronously via FastAPI BackgroundTasks.
- Built the React/TypeScript frontend with Tailwind (authenticated dashboard, analytics visualizations, QR code generation, dark mode), and set up CI/CD with GitHub Actions to deploy the Dockerized app to Render.

Title-Based Playlist Recommendation System with Bias Mitigation

Dec 2025 – Apr 2026

- Developed a machine learning pipeline to generate playlists from user-provided titles using semantic embeddings and similarity-based retrieval over the Spotify Million Playlist Dataset.
- Generated playlist embeddings using Sentence-BERT, applied K-Means clustering to identify thematic groups, and fine-tuned a transformer using Cross-Entropy and Triplet Loss to learn semantic title representations.
- Designed a voting-based recommendation mechanism with LLM-assisted re-ranking to reduce popularity bias, evaluating the pipeline across 58,584 playlists at a mean NDCG@66 of 0.4210 (consistent with the published benchmark range of 0.4156–0.4311).

Hostel Complaint Management System

Jan 2025 – May 2025

- Developed a full-stack web application with backend APIs (Node.js, Express.js, MongoDB) for complaint management, worker assignment, and task retrieval.
- Built dynamic dashboards using React.js enabling assistants to assign workers and workers to view and manage assigned tasks.
- Enabled real-time complaint status tracking across student, assistant, and maintenance worker roles.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL (MySQL), HTML/CSS

Frameworks & Databases: React.js, Node.js, Express.js, FastAPI, Tailwind CSS, MongoDB, PostgreSQL, Redis

Machine Learning: PyTorch, Scikit-learn, Sentence Transformers, HuggingFace Transformers, Pandas, NumPy

Tools & DevOps: Git, GitHub, Docker, GitHub Actions (CI/CD), VS Code, Jupyter Notebook, Google Colab, Eclipse

Problem Solving: Solved 300+ DSA problems on LeetCode, HackerRank, and GeeksforGeeks

Concepts: REST APIs, JWT, bcrypt, Recommender Systems

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

Python for Data Science — NPTEL, Secured a Top 5% Rank in the nationwide certification exam (2025)

College of Engineering, Guindy

Chennai, Tamil Nadu

National Service Scheme (NSS) – Joint Secretary

Jul 2025 – Jun 2026

- Organized a village outreach camp (medical camps, cleaning drives, awareness rally) and led two campus initiatives: "Inaippu" (Aadhar/Voter ID registration) and "Mellinam" (Career Guidance).